

# Alex Fuentes Raventós

XR Developer | Research Assistant | PhD Candidate in Computer Science

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## Profile

XR researcher and developer combining applied R&D at EventLAB with a PhD in Computer Science focused on immersive technologies for health and wellbeing. Specialised in taking XR systems from research prototypes to real-world clinical and institutional use, including an oncology platform designed for multiple hospitals and behavioural interventions being tested in prison settings. Work spans advanced XR engineering, physiological sensing, EEG, and multimodal data pipelines.

## Professional Experience

**EventLAB** **Dec 2023 – Present**  
*XR Developer / Research Assistant*

- Developed an XR oncology platform for exercise and self-observation, designed for use across multiple hospitals and clinical research with oncology patients.
- Developed a traffic-safety and mobile-phone-awareness simulation currently being used by the Department of Justice in testing with participants in prison.
- Build multimodal experimental pipelines combining XR telemetry with physiological/EEG acquisition, Lab Streaming Layer (LSL), and real-time event synchronization.

**Marvut Technologies** **Jun 2023 – Dec 2023**  
*XR Developer*

- Built VR catalogue showrooms in Unity for client companies.
- Developed an orchestra-conducting simulation connecting Unity, Python, and Arduino-based sensors.
- Implemented hand-tracking and hologram interactions using Leap Motion and Kinect.

**Asuni / McNeel Europe** **Jan 2023 – Jun 2023**  
*Software Developer*

- Developed C++ features for BIM plugins for AutoCAD and Rhino, including Lands Design and VisualARQ.

**Muse Scene Lab** **Jan 2023 – Jun 2023**  
*VR Developer Collaborator*

- Developed a real-time orchestra interaction prototype using Unity and hand tracking.

**The MindKind** **Apr 2022 – Jan 2023**  
*Software Engineer / Artificial General Intelligence Researcher*

- Developed Unity environments and agents and implemented Python reinforcement-learning experiments based on sensory inputs.
- Built socket-based communication between Python and C# for agent-environment integration.

## Education

**PhD in Computer Science** **2024 – Present**  
*Universitat Autònoma de Barcelona (UAB) – Research line: Artificial Intelligence*

Research focus: design and validation of integrated extended-reality systems for health and wellbeing.

**MSc in Computer Science – Augmented and Virtual Reality** **2021 – 2022**  
*Trinity College Dublin, Ireland – Distinction*

**Erasmus Programme** **2020 – 2021**  
*Warsaw University of Technology, Poland*

**BSc in Computer Science and Software Engineering** **2016 – 2021**  
*University of Barcelona, Spain*

## Publications and Presentations

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### **NEOBIC-XR - Frontiers in Virtual Reality** 2026

First author of NEOBIC-XR: A Home-Based Extended Reality Exercise Platform and Multicenter Evaluation Framework for Supportive Oncology Care. Methods article describing the platform and multicentre evaluation framework.

### **IEEE ISMAR - accepted poster, upcoming presentation** 2026

First-author poster: Feeling Together Without Talking: Behavioral Signatures of Co-presence in Silent Cooperative VR.

### **CuttingGardens / CuttingEEG, Barcelona - invited talks** 2026

Two talks: Measuring Social Co-presence in Virtual Reality Using EEG, with Ana Zappa, covering preliminary analysis and the post-processing pipeline; and About Time: Precise Event Markers in VR Experiments, with Michael Wiesing.

### **CHI - Conference on Human Factors in Computing Systems** 2026

Co-author and co-organiser of The Co-Living between Reality and Virtuality as a Daily Routine, published in the CHI 2026 Extended Abstracts.

### **Council of Europe - Human Rights in Immersive Realities, Strasbourg** 2026

Demo and poster chair; presented Empowering Oncology Patients Through XR Self-Observation and Mixed Reality Exercise.

### **CIPIE - oncology poster** 2026

Presented The NEOBIC-VR protocol: evaluation of a virtual reality exercise and self-observation program for oncology patients.

### **VR Summit and Council of Europe** 2025

VR Summit invited speaker; attendee and demo presenter at the Council of Europe expert workshop on freedom of expression in immersive realities.

## Research Software

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### **ChronoXR - Unity/OpenXR timing framework** 2026

Developed and validated a framework for stimulus and response timing in XR experiments. Available as an open-source Unity package with integration samples and a ready-to-use project template. With Mel Slater and Michael Wiesing.

## Teaching

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### **Deep Tech Brain - University of Barcelona** 2024+

Coordinator and lecturer for the Virtual Reality module, connecting XR, AI and neuroscience.

## Technical Skills

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Programming: C#, C++, Python; experience with Java, Node.js, Arduino and MongoDB.

XR / Graphics: Unity 3D/XR, OpenXR, OpenGL, Unity AR; Unreal Engine and Photon Networking.

Research / Integration: EEG, physiological sensing, Lab Streaming Layer, event timing, sensor integration, hand tracking, Leap Motion and Kinect.

Tools: Git, GitHub, GitLab; Blender and Maya.

## Languages

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Catalan | Spanish | English (TOEFL iBT: 97, 2021)